Transition to Nftables

Every major distribution in the open source world is moving towards nftables as the default firewall. In short, the venerable Iptables is now dead. This article is a tutorial on how to build nftables.

Currently, there is an *iptables-nft* backend that is compatible with nftables but soon, even this will not be available. Also, as noted by Red Hat developers, sometimes it may translate the rules incorrectly. Rather than rely on an iptables-to-nftables converter, we need to know how to build our own nftables. In nftables, all the address families come under one rule. Nftables runs in the user space unlike iptables, where every module is in the kernel. It also needs less kernel updates and comes with new features such as maps, families and dictionaries.

Address families

Address families determine the types of packets that are processed. There are six address families in nftables and they are:

- ip
- ipv6
- inet
- arp
- bridge
- netdev

In nftables, the ipv4 and ipv6 protocols are combined into one single family called inet. So we do not need to specify two rules – one for ipv4 and another for ipv6. If no address family is specified, it will default to ip protocol, i.e., ipv4. Our area of interest lies in the inet family, since most home users will use either ipv4 or ipv6 protocols (see Figure 1).

Nftables

A typical nftable rule contains three parts – table, chain and rules.

Tables are containers for chains and rules. They are identified by their address families and their names. Chains contain the rules needed for the *inet/arp/bridge/netdev* protocols and are of three types -- *filter, NAT* and *route*. Nftable rules can be loaded from a script or they can be typed into a terminal and then saved as a rule-set. For home users, the default chain will be *filter*. The inet family contains the following hooks:

- Input
- Output
- Forward
- Pre-routing
- Post-routing

To script or not to script?

One of the biggest questions is whether we can use a firewall script or not. The answer is: it's your choice. Here's some advice – if you have hundreds of rules in your firewall, then it is best to use a script, but if you are a typical home user, then you can type the commands in the terminal and then load your rule-set. Each option has its own advantages and disadvantages. In this article, we will type them in the terminal to build our firewall.

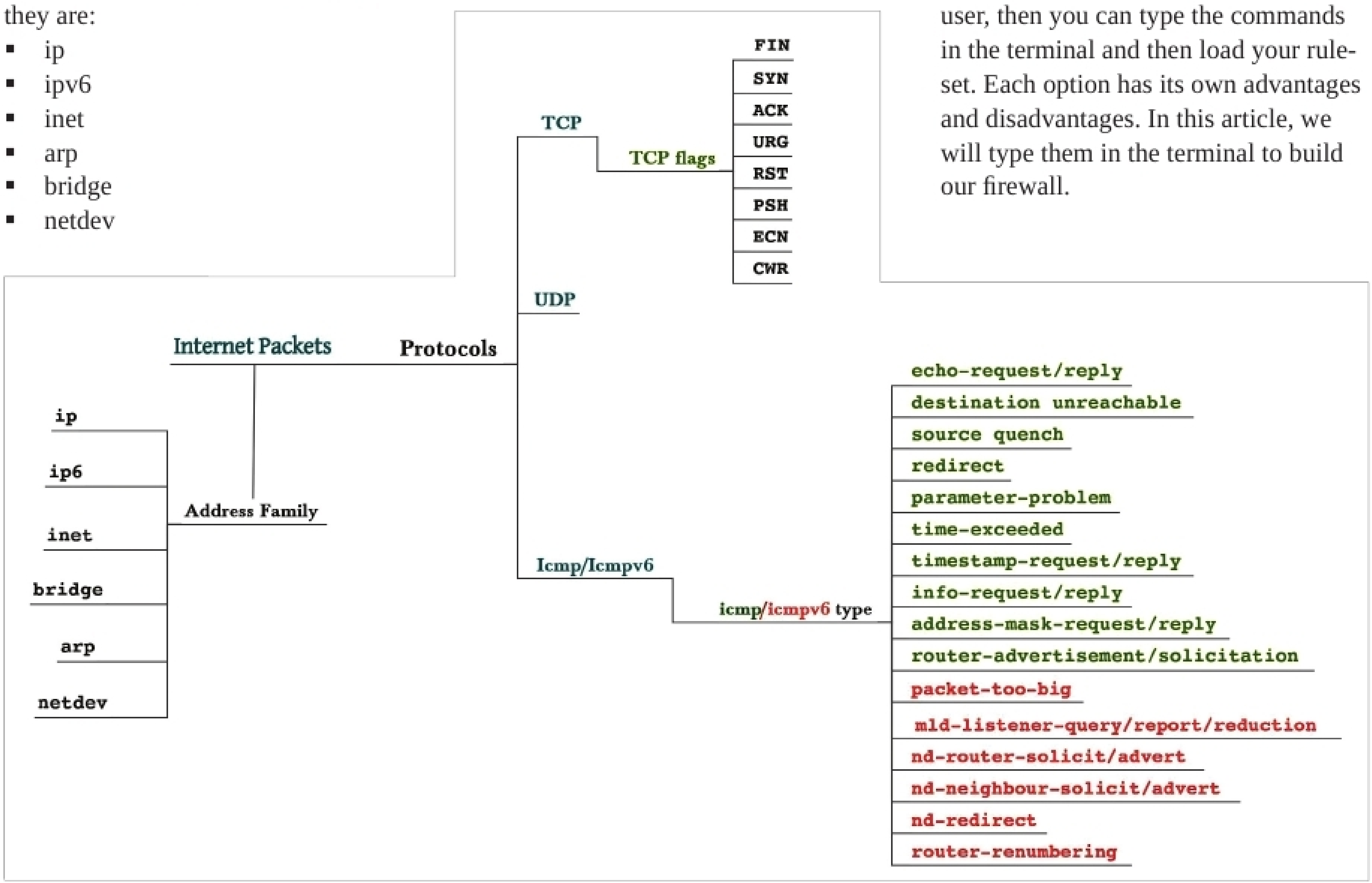

Figure 1: Internet packets